L4: Knowledge Check

1 Fill in the Blank 1 point

If the in-degree distribution of a directed network follows a power law, where $\alpha > 2$, the statements below are FALSE or TRUE?

1. The in-degree cumulative distribution function also follows a power law with the same exponent. ______

2. The in-degree distribution is highly skewed towards values that are much lower than the mean. ______

3. The mean of the in-degree distribution is finite. ______

4. The out-degree degree distribution also follows a power-law distribution. ______

- True
- False
- True
- False
- True
- False
- True
- False

2 Fill in the Blank 1 point

The following statements below are FALSE or TRUE?

1. To check whether a given network follows a power-law distribution, find the degree range between k_sat and k_cut, and check whether it decreases linearly in a log-linear scale plot. ______

2. The low degree saturation k_sat causes a reduction, relative to an ideal power-law form, at low degrees. ______

3. The high degree cutoff k_cut is generated by structural constraints and it causes a deviation from an ideal power-law at very high degree values. ______

4. To check whether a given network follows a power-law distribution, rescale the distribution by k - k_sat, consider the range up to k_cut, and examine if the rescaled distribution decreases linearly in a log-log plot. ______

- True
- False
- True
- False
- True
- False
- True
- False

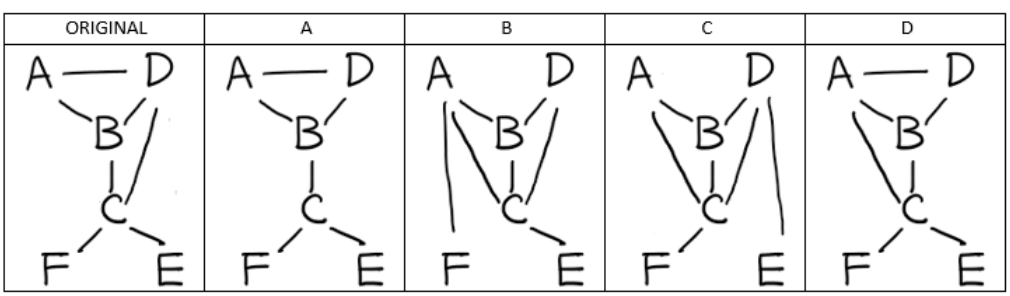
3 Multiple choice 1 point

Let G be a power-law network with 9 nodes. If its minimum degree k_min is 4 and α is 3, find the maximum degree k_max of G.

- ○ 12
- ○ 13
- ○ 15
- ○ 16

4 Multiple choice 1 point

The leftmost graph below is the original network. Which one of the graphs at the right side can be categorized as degree-preserving randomization?



- ○ A
- ○ B
- ○ C
- ○ D

5 Fill in the Blank 1 point

The following statements about the Configuration Model are FALSE or TRUE?

1. It can produce power-law networks for any value of $\alpha>2$. ______

2. It may produce a graph with self-loops and/or multi-edges. ______

3. The only requirement is that the node degrees are non-negative integers. ______

4. It can generate a network with any degree sequence. ______

- True
- False
- True
- False
- True
- False
- True
- False

6 Multiple choice 1 point

In the link selection model, what is the relationship between the degree of an existing node v and the probability that a new node connects to v?

- Exponential
- Linear
- Quadratic
- No relationship between the two